SIMATS ENGINEERING

ASSESSMENT TOOL II — Visualization Analysis Report

Course: Data Handling and Visualization	**Code: DSA06**	**CO: CO5**
Duration: 180 min	**Total Marks: 100**	**Reg No : 192324298** **Name: Akash S**

COURSE OUTCOME COVERED

CO5: Analyze and evaluate visualization choices to communicate patterns, comparisons, relationships, and potential misleading effects through appropriate visualization methods. (BL4)

Activity Tool : Visualization Analysis Report

Weightage: 20%

Code of Conduct:

I Akash S 19234298 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ____________________

Assessment Criteria

Criteria	Visualization Selection and Understanding 25%	Visualization Analysis and Interpretation 25%	Application of Visualization Principles 20%	Organization and Presentation 20%	Accuracy 10%	Clarity 5%	Total
Q1							
Q2							
Q3							
Q4							
Q5							

Visualization Analysis Report – Assessment Tool

R Programming Practical Questions with Datasets

Instructions: Answer all five questions. Use R programming to construct the required visualizations, analyze the patterns, compare visualization choices where requested, and identify potential visualization pitfalls. Use the given data exactly unless a transformation is explicitly requested.

Q1. Comparing Distributions – Histogram, Density and ECDF

A quality-control team records the following processing times (in minutes) for 20 randomly selected tasks. Using R, create a histogram and a density plot for the data. Also construct an empirical cumulative distribution function (ECDF). Compare the three visualizations and explain which aspects of the distribution—center, spread, skewness, and cumulative behavior—are communicated most clearly by each.

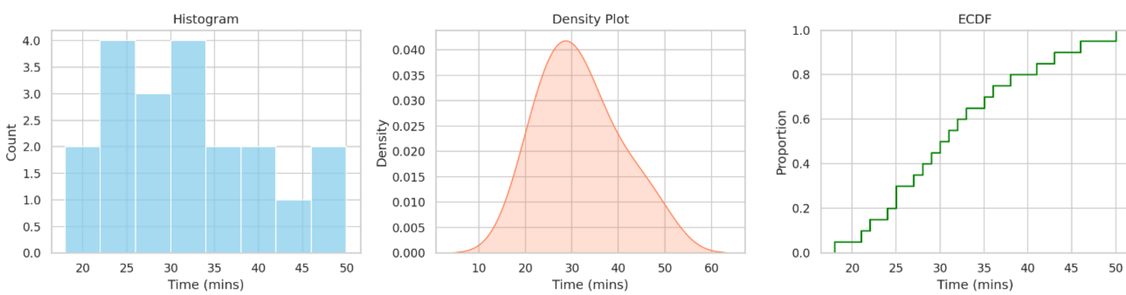
Data: Time = 18, 21, 22, 24, 25, 25, 27, 28, 29, 30, 31, 32, 33, 35, 36, 38, 41, 43, 46, 50

Expected R tasks: histogram, density estimation, ECDF, appropriate labels/scales, comparison and interpretation.

Answer:

R Code Used:

```
Time <- c(18, 21, 22, 24, 25, 25, 27, 28, 29, 30, 31, 32, 33, 35, 36, 38, 41, 43, 46, 50)
par(mfrow=c(1,3))
hist(Time, main="Histogram", col="skyblue")
plot(density(Time), main="Density Plot", col="coral", lwd=2)
plot(ecdf(Time), main="ECDF", col="green", lwd=2)
```



Interpretation:
1. Histogram: Best for showing discrete frequency counts and the general shape of the data bins. It easily communicates that the bulk of processing times lie between 20 and 40 minutes.
2. Density Plot: Provides a smoothed, continuous view of the distribution. It highlights a slight right-skewness in the data spread that is somewhat obscured by the histogram bins.
3. ECDF: Best for answering cumulative probability and percentile questions. We can clearly see that about 50% of the tasks are completed in under 30 minutes.

Q2. Q-Q Plot and Normality Analysis

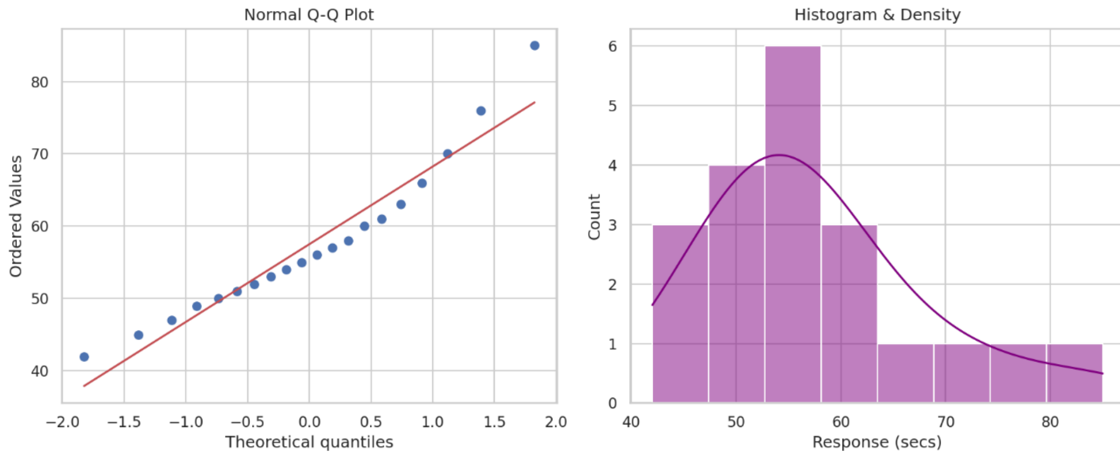
A researcher measures the following response times (in seconds). Use R to create a normal Q-Q plot and a histogram/density visualization. Assess whether the observations are reasonably consistent with a normal distribution. Identify any departures from the reference line and explain what they indicate about the distribution.

Data: Response = 42, 45, 47, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 60, 61, 63, 66, 70, 76, 85

Expected R tasks: qqnorm()/qqline() or ggplot2 Q-Q plot, histogram/density plot, interpretation of deviations and outliers.

Answer:

```
Response <- c(42, 45, 47, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 60, 61, 63, 66, 70, 76, 85)
par(mfrow=c(1,2))
qqnorm(Response); qqline(Response, col="red")
hist(Response, prob=TRUE); lines(density(Response), col="purple", lwd=2)
```



Interpretation:
The observations deviate from a perfect normal distribution. The Q-Q plot exhibits a significant upward departure from the red reference line at the upper tail. This indicates that the higher response times (e.g., 76, 85 seconds) are much larger than what would be expected under a normal distribution. The histogram and density plot confirm this by showing a distinct right skew (positive skewness) with a long tail stretching towards the higher values. Therefore, the data is not strictly normal.

Q3. Visualizing Many Distributions at Once

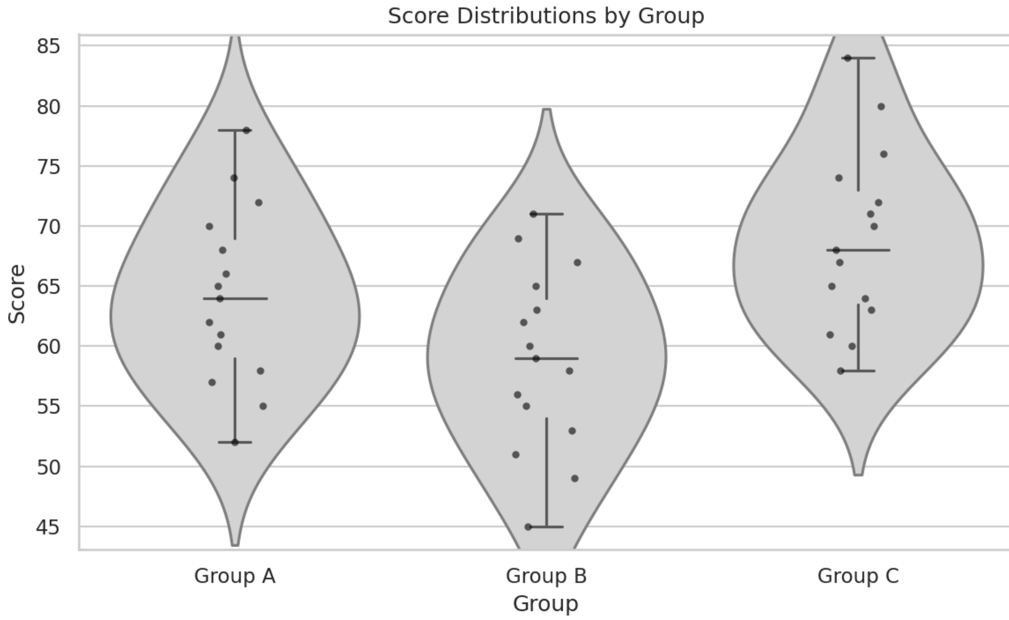
Three groups of students have examination scores recorded below. Create a suitable visualization that compares the distributions of all three groups at once. Use boxplots and/or violin plots, and optionally overlay individual observations. Analyze differences in center, spread, and possible outliers. Explain why the selected visualization is more informative than displaying three separate plots.

Group A = 52,55,57,58,60,61,62,64,65,66,68,70,72,74,78;
Group B = 45,49,51,53,55,56,58,59,60,62,63,65,67,69,71;
Group C = 58,60,61,63,64,65,67,68,70,71,72,74,76,80,84

Expected R tasks: data reshaping, grouped boxplot/violin plot, optional jitter, comparison of distributions, interpretation.

Answer:

```
library(ggplot2)
# Data reshaping
scores <- data.frame(
  Score = c(52,55,57,58,60,61,62,64,65,66,68,70,72,74,78,
           45,49,51,53,55,56,58,59,60,62,63,65,67,69,71,
           58,60,61,63,64,65,67,68,70,71,72,74,76,80,84),
  Group = rep(c("Group A", "Group B", "Group C"), each=15)
)
ggplot(scores, aes(x=Group, y=Score, fill=Group)) +
  geom_violin(alpha=0.3) + geom_boxplot(width=0.2) +
  geom_jitter(width=0.1, alpha=0.6) + theme_minimal()
```



Interpretation:
By overlaying a boxplot, violin plot, and individual points (jitter), we capture the center, spread, and raw data simultaneously.
- **Center:** Group C has the highest median score (~68), while Group B has the lowest (~59).
- **Spread:** Group A and C have relatively similar, wider spreads compared to Group B, which is slightly more concentrated.
- **Why this is better:** Plotting them on a single axis allows for immediate side-by-side benchmarking. Three separate plots would require the reader to constantly shift visual focus and manually map the y-axes, which drastically impairs comparison speed and accuracy.

Q4. Relationships and Multivariate Visualization

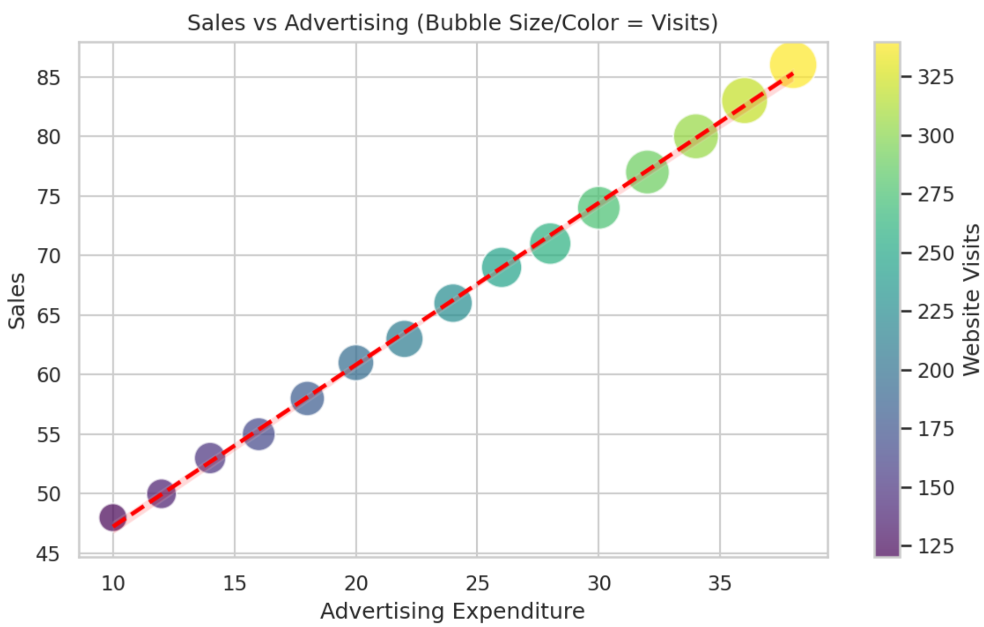
A study records advertising expenditure, website visits, and sales for 15 observations. Create a scatterplot of advertising expenditure versus sales and use an appropriate visual encoding to incorporate website visits. Add a fitted trend line. Analyze the strength and direction of the relationship and explain whether the additional variable improves the interpretation or risks cluttering the visualization.

Advertising = 10,12,14,16,18,20,22,24,26,28,30,32,34,36,38;
Visits = 120,135,150,165,180,195,210,225,245,260,275,290,305,320,340;
Sales = 48,50,53,55,58,61,63,66,69,71,74,77,80,83,86

Expected R tasks: scatterplot, trend line, third-variable encoding, appropriate legend/scale, relationship analysis and critique.

Answer:

```
library(ggplot2)
df <- data.frame(
  Advertising = c(10,12,14,16,18,20,22,24,26,28,30,32,34,36,38),
  Visits = c(120,135,150,165,180,195,210,225,245,260,275,290,305,320,340),
  Sales = c(48,50,53,55,58,61,63,66,69,71,74,77,80,83,86)
)
ggplot(df, aes(x=Advertising, y=Sales, size=Visits, color=Visits)) +
  geom_point(alpha=0.7) + geom_smooth(method="lm", se=FALSE, color="red") +
  theme_minimal() + labs(title="Sales vs Advertising (Size/Color = Visits)")
```



Interpretation:
The trend line demonstrates a strong, positive linear relationship between Advertising Expenditure and Sales. The additional variable, "Website Visits", is effectively encoded using point size and color (bubble chart). This mapping shows that as advertising increases, website visits also increase proportionally, leading to higher sales.

Because the third variable aligns strongly with the primary trend without causing extreme overlap, it improves the interpretation by revealing a likely causal pathway (Ads -> Visits -> Sales) without significantly cluttering the visualization.

Q5. Misleading Visualization and Redesign

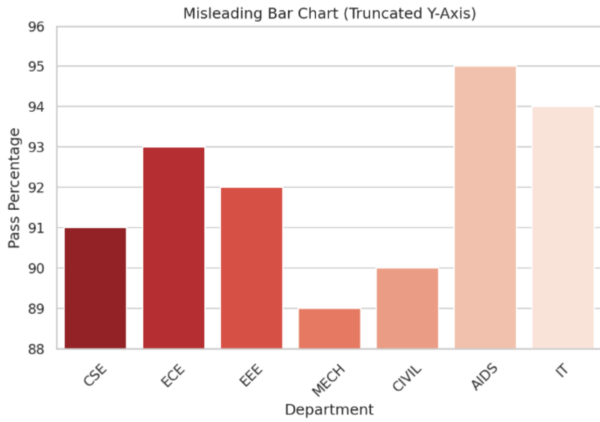
A college reports the following pass percentages for seven departments. Create two visualizations: (1) a misleading version using a truncated y-axis and an inappropriate visual emphasis, and (2) a redesigned version using an appropriate baseline, ordered categories, and clear labels. Compare the two plots and explain how the first visualization can exaggerate differences and why the redesigned visualization communicates the data more honestly.

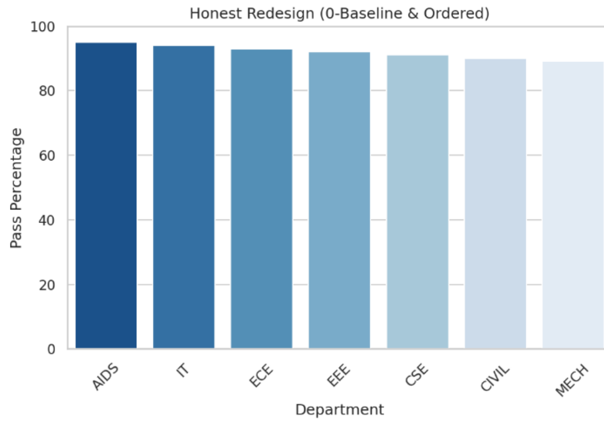
Department = CSE, ECE, EEE, MECH, CIVIL, AIDS, IT;
Pass_Percentage = 91,93,92,89,90,95,94

Expected R tasks: bar chart with controlled y-axis, redesigned bar chart, ordering, labels, misleading-axis critique, accessibility and communication analysis.

Answer:

```
# Misleading Plot
barplot(Pass_Percentage, names.arg=Department, ylim=c(88, 96),
      col="red", main="Misleading (Truncated Axis)")
# Honest Redesign
df_pass <- data.frame(Department, Pass_Percentage)
df_pass <- df_pass[order(-df_pass$Pass_Percentage), ]
barplot(df_pass$Pass_Percentage, names.arg=df_pass$Department, ylim=c(0, 100),
      col="steelblue", main="Honest Redesign (0-Baseline & Ordered)")
```





Interpretation:

The misleading plot (left) truncates the y-axis (starting at 88 instead of 0). This creates an artificial visual disparity—for instance, making AIDS (95%) look many times better than MECH (89%), despite the absolute difference being only 6%. It sensationalizes minor variations.

The redesigned plot (right) corrects this by implementing a 0-100 baseline and sorting the bars in descending order. This provides an honest, proportional view of the data, accurately showing that all departments have very high and quite similar pass rates. Ordering the categories further aids readability and direct comparison.